SG98 Headless Controller Remote Safety Simulation

Communication corner-case review for the headless controller after adding low-rate remote heartbeat, controller-side deadman stop, and valid-command ESP-NOW boot recovery.

Headless only
Source: host-side workflow simulation

HEARTBEAT
1 second
Remote sends while controller is in a running state.

DEADMAN
7.5 seconds
Headless controller stops running states with no valid activity.

BOOT RECOVERY
60 seconds
Saved ESP-NOW falls back to Web recovery if no valid command arrives.

SCOPE
Headless only
Full-size controllers are excluded by local UI capability checks.

R–>C means remote, ESP-NOW peer, or remote-control sender to controller.

C–>R means controller response, status, ack, or discovery reply to remote.

Motor-safe result: no simulated case leaves a headless running motor uncontrolled after remote traffic disappears.

#	Case	Transport	Comm / Disconnection Direction	Final Controller State	Motor Off	Recovery	Result	Notes
1	**ESP-NOW boot, no remote**	ESP-NOW saved	No R–>C traffic	**IDLE** Web recovery	n/a	60s	OK	No valid command seen; opens Web recovery at 60s.
2	**ESP-NOW boot, stray `DISCOVER` only**	ESP-NOW saved	R–>C `DISCOVER`; C–>R discovery response	**IDLE** Web recovery	n/a	60s	OK	`DISCOVER` does not satisfy recovery anymore.
3	**ESP-NOW boot, valid remote command**	ESP-NOW saved	R–>C OK; C–>R OK	**IDLE** ESP-NOW stays	8.0s	-	OK	Valid command satisfies boot recovery.
4	**`START` delivered, remote power loss**	ESP-NOW / Web	Both directions down after `START`	**IDLE** Stopped, link down	8.6s	-	OK	Deadman stops at about last activity plus timeout.
5	**`START` delivered, remote transmit lost**	ESP-NOW / Web	R–>C down; C–>R may still transmit	**IDLE** Stopped	8.6s	-	OK	Heartbeats cannot reach controller, so deadman stops.
6	**`START` delivered, controller response lost**	ESP-NOW / Web	C–>R down; R–>C alive	**GRINDING** Controller still controlled	n/a	-	OK	No timeout because remote heartbeat still arrives. Remote UI may look stale.
7	**`START` command lost**	ESP-NOW / Web	R–>C down during `START`	**IDLE** Idle	n/a	-	OK	Nothing starts.
8	**`STOP` command lost after running**	ESP-NOW / Web	R–>C down before `STOP`	**IDLE** Stopped	10.6s	-	OK	Deadman is the backstop for a lost `STOP`.
9	**`STOP` delivered, response lost**	ESP-NOW / Web	R–>C OK; C–>R down	**IDLE** Idle	4.0s	-	OK	Controller stops immediately; remote can retry/status-recover later.
10	**`STOP` delivered with button-purge enabled**	ESP-NOW / Web	R–>C OK; C–>R may be down	**IDLE** Idle after purge	7.0s	-	OK	Expected purge tail before idle.
11	**Running state entered without remote traffic**	Web / ESP-NOW	No traffic after state entry	**IDLE** Stopped	8.6s	-	OK	Running-observed timestamp still trips the deadman.

#	Case	Transport	Comm / Disconnection Direction	Final Controller State	Motor Off	Recovery	Result	Notes
12	**Web UI active while remote path lost**	Web	Remote R–>C down; Web API to controller alive	**GRINDING** Still running by Web control	n/a	-	OK	Web UI traffic intentionally keeps headless control alive.
13	**GbW run, remote loss**	ESP-NOW / Web	Both directions down after START	**IDLE_GBW** Stopped in GbW idle	8.6s	-	OK	Deadman returns to IDLE_GBW.
14	**Remote seen once, then lost while idle**	ESP-NOW saved	One valid R–>C, then none	**IDLE** ESP-NOW only	n/a	-	WARN	Motor-safe. If reachability is needed, power-cycle returns to boot recovery.

Safety Conclusion

For headless operation, the low-rate heartbeat plus controller-side deadman closes the uncontrolled running cases. If remote traffic disappears while grinding, purging, or GbW grinding, the controller transitions back to an idle state and sets motor command to zero.

Reachability Note

The remaining warning is idle-only: after one valid ESP-NOW command, a later idle remote loss can leave the controller ESP-NOW-only. That is motor-safe, and power-cycling the headless controller re-enters the 60s Web recovery path if no valid remote command arrives.